

Quiz 2 2026
Biophysics(BIO215)

Time: 45 minutes

Total Marks:20

Section-A (MCQ 1*10 Marks)

1) In a theoretical system with two energy levels (E1 and E2), the distribution of particles is governed by the Boltzmann relationship. Under which of the following physical conditions would you observe an equal number of particles in both the ground state and the excited state?

- A) The system is at absolute zero.
- B) All particles are in the ground state.
- C) The energy gap ΔE is zero or the temperature is effectively infinite.**
- D) The system has reached a state of maximum potential energy.

2) Which physical principle underlies the derivation of the Boltzmann distribution?

- (A) Change in free energy
- (B) Maximization of Entropy**
- (C) Change in Enthalpy
- (D) Newton's Third Law

3) In a sci-fi drug-design mission, a nanoligand is sent to bind a mutant viral protein inside an alien cell. What is the primary goal of molecular docking in this scenario

- A. To observe the folding of nanoligand inside the protein's active site.
- B. To predict the ligand's orientation and stable binding pose in the receptor active site**
- C. To sequence the alien genome as seq of nanoligand is already known
- D. To convert the receptor into RNA via nanoligand induced reverse translation

4) A space-biology lab has no clue where the binding pocket is on a newly discovered protein. Which docking strategy is most appropriate?

- A. Active-site docking with multiple poses of ligand followed by evaluation of docked poses.
- B. Blind docking**
- C. Inverse docking
- D. Rigid docking followed by molecular dynamics to identify stable binding poses and sites

5) A ligand must adapt its shape while the receptor side chains also shift slightly during docking. Which model is being used?

- A. Rigid docking
- B. Flexible docking**
- C. Virtual screening only
- D. Inverse docking

6) In the thermodynamic worldview, entropy is best interpreted as:

- A. Energy divided by charge
- B. A simple logarithm of the number of accessible states**
- C. The exact same thing as temperature
- D. A measure of bond length

7) A robot scientist wants to screen thousands of candidate molecules quickly before high-accuracy docking & refinement. Which docking strategy type best matches that need?

- A. Virtual screening -> Rigid docking**
- B. Virtual screening -> Flexible docking
- C. Target Protein-Complement protein docking -> Virtual screening
- D. Virtual screening -> Flexible docking -> Rigid docking

8) Which pairing is correctly matched with its interaction type?

- A. AutoDock Vina — protein-protein docking
- B. HADDOCK — protein-protein docking**
- C. rDock — protein-protein docking
- D. ZDOCK — protein-ligand docking

9) A ligand is docked against a library of many proteins to detect hidden off-targets in a futuristic therapy. This is called:

- A. Blind docking
- B. Single ligand docking
- C. Inverse docking**
- D. Active-site docking

10) A molecule in a biological system moves from a low-energy gas molecule to occupy two chambers inside a container. Chamber A has a volume of 2 L, and Chamber B has a volume of 6 L. The energy of the molecule is the same in both chambers.

Where is the molecule more likely to be found at equilibrium?

- A. Chamber A, because smaller volume increases confinement
- B. Chamber B, because larger volume provides more accessible microstates**

- C. Equal probability in both chambers because energy is same
- D. Cannot be determined without temperature

Section-B (2*5=10 mark)

1.) A system exists in thermal equilibrium at temperature T and can occupy two energy states:

- **State A: Energy $E_A=0$**
- **State B: Energy $E_B=5kT$**

a. Using the Boltzmann distribution, calculate the ratio of probabilities: P_B/P_A

b. Interpret the result in terms of population of the two states.

Ans a .

$$\frac{P_B}{P_A} = e^{-(E_B - E_A)/kT}$$

$$\frac{P_B}{P_A} = e^{-(5kT - 0)/kT} = e^{-5}$$

$$\frac{P_B}{P_A} \approx 0.0067$$

Ans b. State B is much less populated than State A. Only about 0.67% of particles occupy the higher energy state. This shows that higher energy states are exponentially less probable

2.) A researcher is performing flexible docking of a novel drug candidate against a protein receptor. After running the simulation, they obtain several possible binding poses, but the software identifies one specific pose as the "most stable".

Part A: Based on the thermodynamic goals of molecular docking, what specific physical value determines which pose is the most stable, and should this value be high or low? (1 Mark)

Part B: If the researcher finds that the ligand binds effectively to the known active site but also binds to a different, unintended protein in a separate inverse docking experiment, what is the clinical significance of this second finding? (1 Mark)

Ans: a. The stability is determined by the binding affinity (or Gibbs free energy, ΔG). The value should be low (more negative), as a more negative binding affinity signifies more favorable and stable binding.

b. This signifies a potential off-target effect. In a clinical context, this suggests the drug candidate might cause side effects or toxicity by interacting with proteins it was not intended to target.

3.) A researcher is preparing a protein receptor (PDB ID: 3DTC) for a docking study. The process involves multiple computational modifications before the docking can begin.

Part a: List any four specific structural "cleanup" steps that must be performed on the raw PDB file before it is ready for docking. (1 Mark)

Part b: Why is the final receptor file saved in the .PDBQT format instead of the standard .PDB format? (1 Mark)

Ans 3. Part a: (Any four of the following):

- Removal of water molecules.
- Deletion of unwanted heteroatoms/existing ligands.
- Repairing or checking for missing atoms.
- Adding polar hydrogen atoms.
- Assigning Kollman charges

Part b: The .PDBQT format is necessary because, unlike a standard PDB file, it includes Partial Charges (Q) and Atom Types (T). These additional parameters are required for the docking algorithm to calculate the electrostatic and van der Waals interactions between the ligand and receptor.

4). A protein is being studied under two experimental conditions:

- (i) inside a rigid sealed chamber where the volume remains fixed
- (ii) inside an open test tube exposed to the atmosphere where the pressure remains constant

For each condition, mention which free energy should be minimized to predict the most stable state of the protein and write its expression.

Answer: (i) For the rigid sealed chamber (constant volume), the system minimizes Helmholtz free energy- $(F=E-TS)$.

(ii) For the open test tube at atmospheric pressure (constant pressure), the system minimizes Gibbs free energy- $(F=H-TS)$

5.) According to the Boltzmann principle, a high-temperature environment naturally promotes molecular disorder. If a biological organism manages to maintain a highly ordered structure in such an environment, how would you explain this using the concepts of entropy and microstates?

Ans 5: Highly ordered structures in a high-temperature environment suggest that the organism is maintaining a smaller set of accessible microstates than expected. Since entropy is essentially the log of the number of states, a very ordered structure corresponds to reduced accessible states, even if the environment would normally favor disorder.